SECTION B

Option A – Alternating Currents

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
7	(a)	(i)	Choosing correct equation (ωBAN) (1) Correct answer = 421 [V] (1)	1	1		2	1	
		(ii)	Choosing correct equation ($BAN\cos\omega t$) (1) Answer = 0 **or** close to 0 (1) Award 1 mark for answer of 1.34	1	1		2	1	
		(iii)	421 [V] **ecf** from (i)		1		1	1	
		(iv)	Period is 20 ms so 5 ms quarter cycle (1) ωt is 90º or equivalent (1) Expect max emf and zero flux (1) **Alternative:** ωt is 90º or equivalent (1) cos 90º = 0 and sin 90º = 1 (1) Expect max emf and zero flux (1)			3	3		
	(b)	(i)	Choosing correct equation i.e. $\frac{1}{2\pi}\sqrt{\frac{1}{LC}}$ (1) Answer = 103 k[Hz] (1)	1	1		2	1	

Question			Marking details	Marks available AO1	AO2	AO3	Total	Maths	Prac
		(ii)	Both reactances cancel **or** total impedance is R **or** equivalent (1) $I = \frac{V}{Z} = \frac{V}{R}$ (or equivalent) (1)	1	1		2		
		(iii)	Any correct reactance (X_C = 10 105, X_L = 10 556) (1) Impedance correctly obtained 485 [Ω] (1) Substitution into $I = \frac{V}{Z}$ **ecf** on Z (1) Correct answer = 5.15 m[A] **ecf** on Z (1)	1	1 1 1		4	4	
		(iv)	Large decrease in current or $\frac{\Delta I}{\Delta f}$ (1) For small increase in frequency or is large (1)	1	1		2	2	
		(v)	Realising ω also changes **OR** using / deriving $\frac{1}{R}\sqrt{\frac{L}{C}}$ (1) Correctly linking valid equation to obtain yes **OR** simply yes e.g.in $\frac{\omega L}{R}$ ω decreases so yes (1)			2	2		
			Question 7 total	**6**	**9**	**5**	**20**	**10**	**0**